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CADRE: Card-Agnostic Domain-Aligned RF Embeddings for Virtual PIN Pads on Passive NFC Cards

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Background & Motivation

The Vulnerability of Digital-Only Authentication



- **Near Field Communication (NFC)** is ubiquitous due to

Why Reader-Side PIN Pads Are NOT a Complete Security Solution?

Reader-side PIN pads



	Standard NFC	NFC + Reader Active PIN Pad (Flawed Fix)	CADRE (Proposed)
Authenticates User?	No	Yes	**Yes**
Reader hardware Cost	Low	High	**Low**
User Hygiene / Shared Surfaces	Safe	Unsafe	**Safe**
Battery Required?	No	No	**No**
Vulnerable to Thermal/Camera Attacks	No	Yes	**No**

- Expensive hardware
- Increased system complexity
- Not scalable for low-cost NFC

Security risks

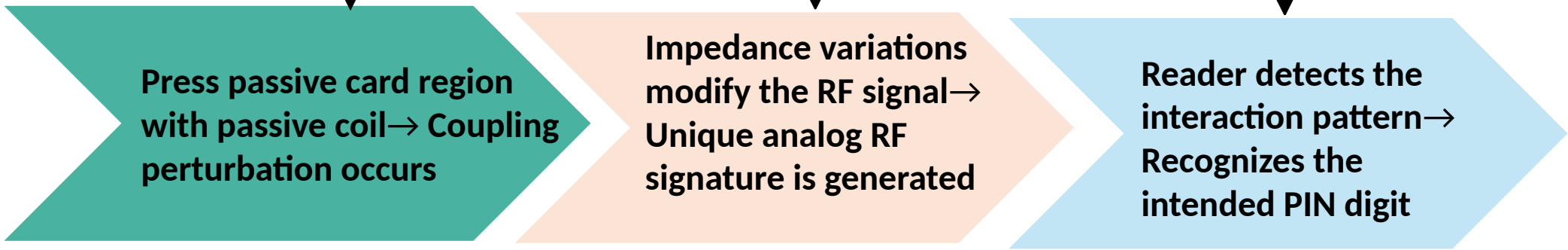


Heat dissipates slowly. A thermal camera can perfectly reconstruct PIN digits up to 30 seconds after the user has walked away.

- Heat signatures persist seconds after touch
- Oils reflect light → visible patterns
- Attacker doesn't need RF knowledge → just a camera
- Thermal attacks → recover pressed digits
- Visual residue (oils) → fingerprint-like patterns

Core Insight

Physical-layer RF signatures contain interaction-specific analog patterns. The card itself becomes the input interface.



Implication: By treating designated regions of a passive NFC card as virtual button press areas, user interactions can be recognized directly through RF signal perturbations without batteries, or active electronics.

How it Works: Mathematical Foundation

Coupling Model:

- $i = \{1,2,p\}$: coil index for reader, card, and button coil.
- L_i: self-inductance of coil i
- M_{iq}: mutual inductance between any coils i and q
- Z_2: NFC card antenna impedance.
- Z_p: passive button-coil impedance.

$$M_{iq} = \mu_0 \pi n_i n_q a_i a_q \int_0^\infty J_0(sp) J_1(sa_i) J_1(sa_q) e^{-s|z_i - z_q|} ds$$

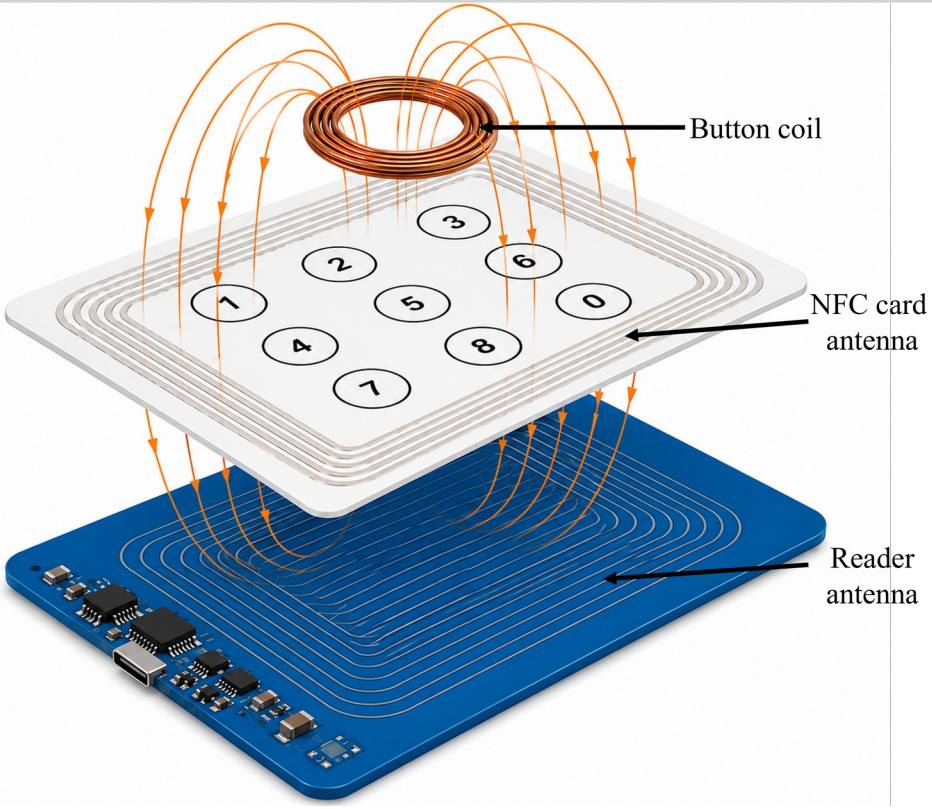
Reflected Impedance:

$$Z_{\text{ref}} = \frac{\omega^2 M_{12}^2}{Z_2} + \frac{\omega^2 M_{1p}^2}{Z_p}$$

- First term: normal reader-card coupling.
- Second term: extra induced-impedance from the button coil.

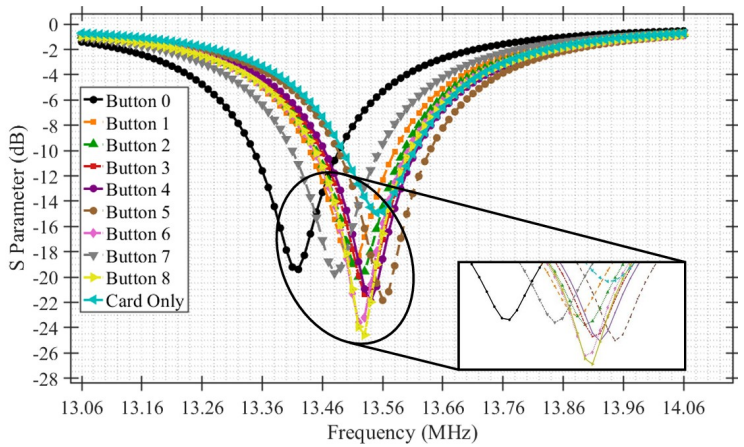
Received RF Signal:

$$S(f) = H(f) k \left[A_{\text{CW}} \delta(f) + A_{\text{card}} \left(\delta(f - f_s) + \delta(f + f_s) \right) \right]$$

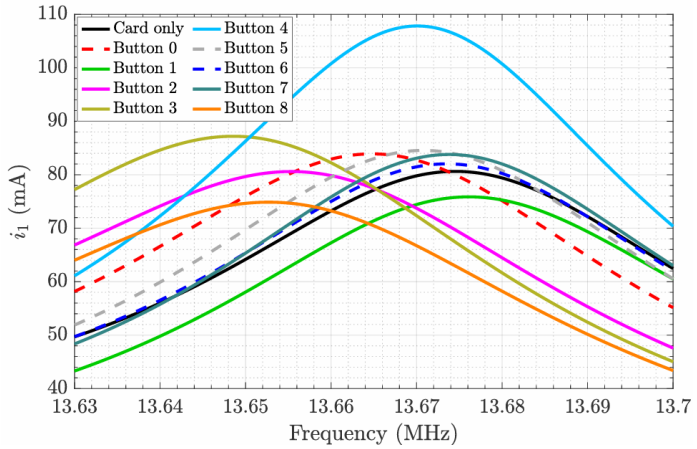


- Each button position changes the coupling path between the button coil and the card antenna, i.e., every press physically alters the mutual inductance M_{1q}, M_{2q} and $k = {}^{M_{1p}}/_{\sqrt{L_1 L_p}}$ without detuning the card's primary resonance.
- This creates a location-dependent RF signature that the reader can observe.
- Although the button coil changes the reflected signal, the main NFC resonance remains stable, so normal card communication is preserved.

Simulation Results



Impact of button coil on the reflection coefficient.



Effect of button coil-induced coupling on reader current variation.

1. Reflection Coefficient: We model the button-coil for 9 predefined positions on NFC card in COMSOL Multiphysics, where distinct S11 resonance shifts show that each button produces subtle yet distinct variations

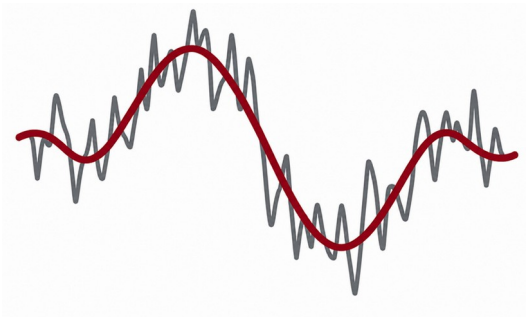
- **Impact:** Shows that passive button interactions can be differentiated without detuning primary resonance **(13-14 MHz).**

2. Reader Current Variation: We model the reader-card-button coupling in MATLAB, where different button locations produce measurable reader-current changes near resonance.

- **Impact:** Demonstrates that button presses generate stable and detectable analog RF signatures that can be captured directly from the reader side.

Two Major Challenges

1



Hardware-Dependent Distortions

- The RF signals produced by a press are short, high-frequency bursts.
- Subtle differences between reflected RF signals are easily obscured by reader noise and environmental factors.

2



Cross-Card Domain Shift

- Every NFC card has slight physical variations (coil geometry, parasitics, carrier detuning, etc.).
- An identical button press on Card A produces a completely different analog signal on Card B.

Result: A standard machine learning model trained on one card will fail on unseen cards.

Proposed Framework: CADRE

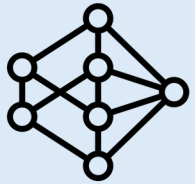
Card-Agnostic Domain-Aligned RF Embeddings

Input RF Signal



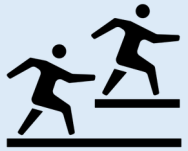
- Short temporal sequences of reflected RF signals are collected from the passive NFC interaction.

Temporal Neural Encoder



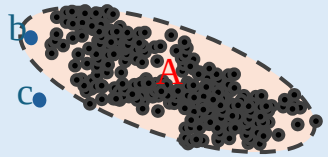
- A temporal deep learning encoder extracts interaction-specific analog RF embeddings associated with each virtual press.

Domain-Adversarial Training



- A domain-alignment module suppresses card-specific hardware distortions and invariant device characteristics, enabling generalization across different NFC cards.

Mahalanobis Recognition



- A distance-aware recognition framework verifies whether the observed signal belongs to a valid class while safely rejecting noise, uncertainty, and spoofed inputs.

Key Goal: Learn RF representations that preserve user interaction information while remaining invariant to card-specific physical differences.

Datasets

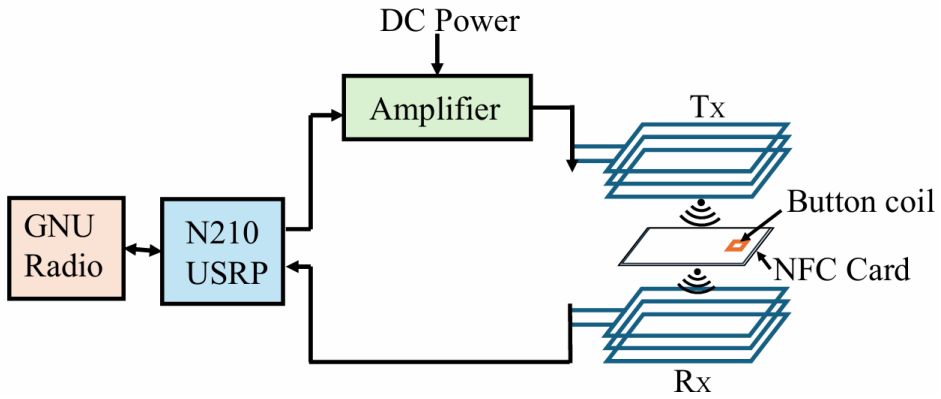
Training and Evaluation Datasets

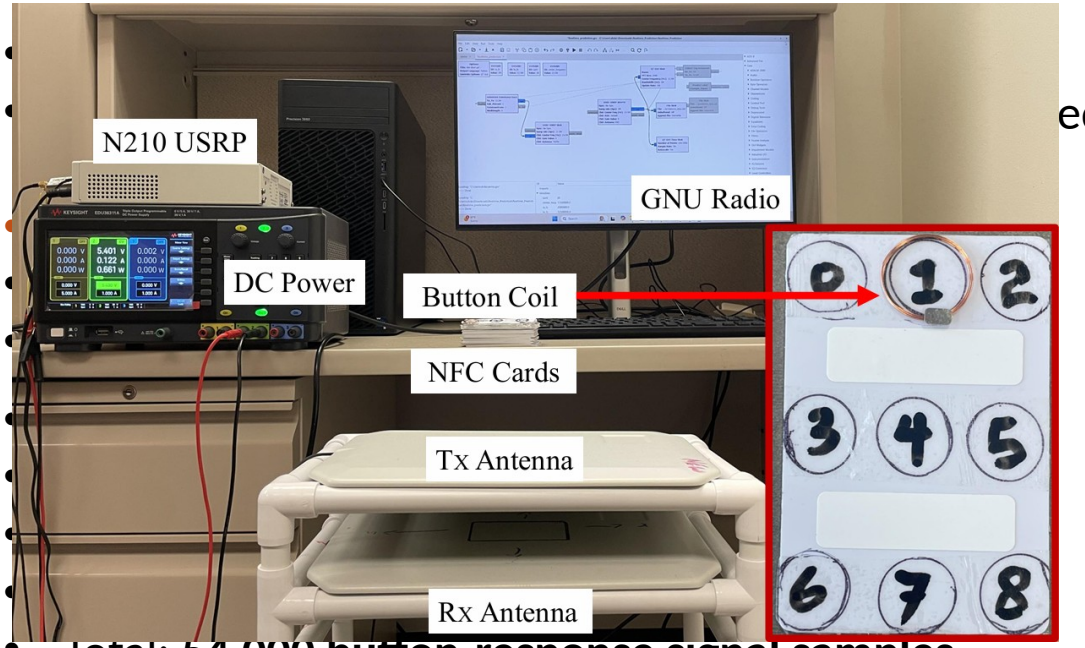
1. Training Dataset:

2. Evaluation Dataset:

- **Condition:** RF response signals are collected across 9 virtual button regions under different button-coil orientations and reader-card coupling conditions.

Data collection experiment setup.





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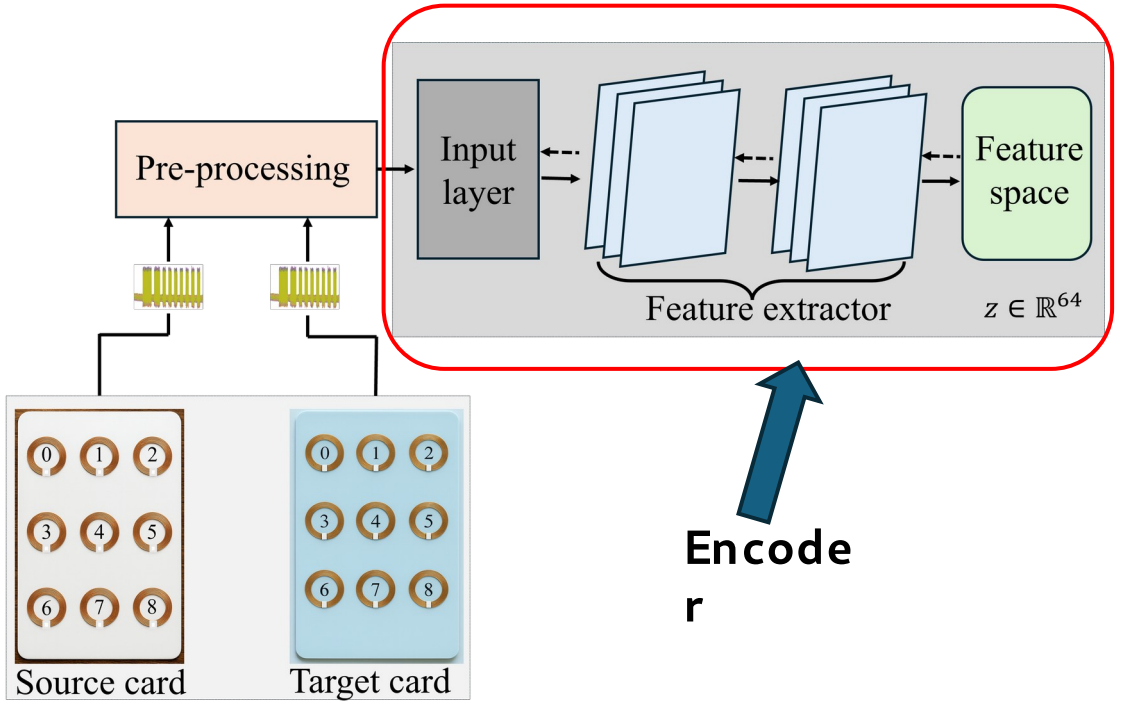
- Total: **54,000 button-response signal samples**

Method 1: Temporal Neural Encoder

The Challenge: Reflected RF response signals are high-dimensional and complex.

The Architecture: A hybrid deep-learning model distills the signal into a compact 64-dimensional feature vector.

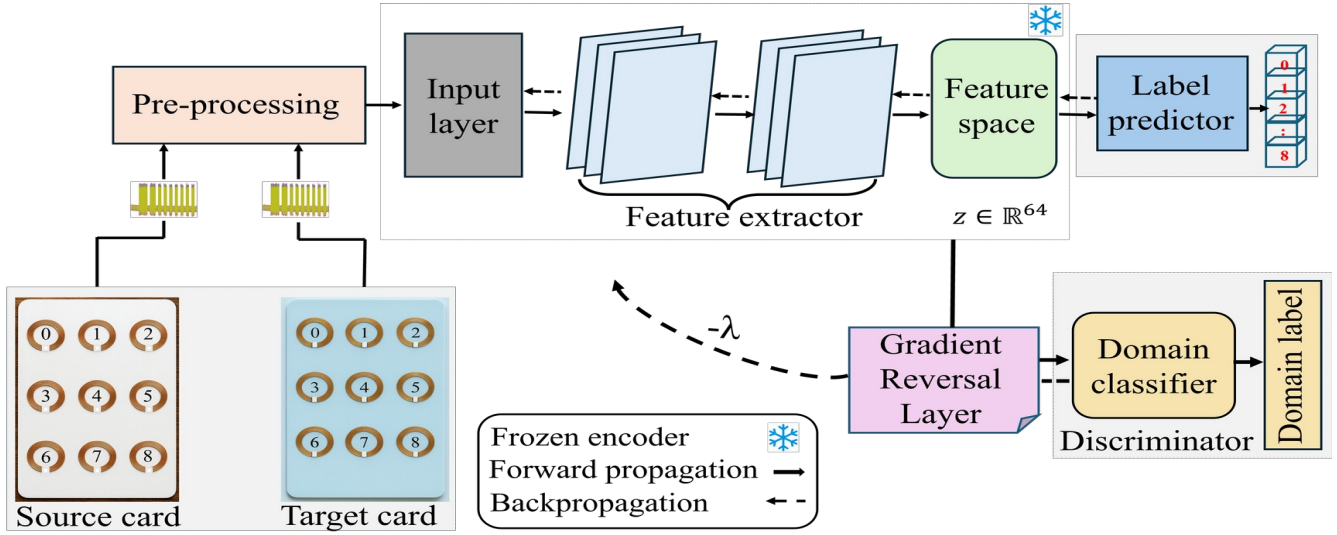
- **CNN Layers:** Two 1D convolutional layers (32 and 64 filters) extract local RF patterns such as amplitude variations, transient peaks, and short-duration perturbations.
- **Normalization & Pooling:** Batch normalization stabilizes feature learning, while max pooling suppresses noise and reduces temporal dimensionality.
- **LSTM Layer:** A 64-unit LSTM models the temporal evolution of the RF response and captures long-term sequential interaction behavior.
- **Output Embedding:** Global average pooling and a dense projection layer compress the learned features into a compact 64-dimensional RF embedding representing the virtual button press.



The Goal: The encoder learns the unique RF patterns needed to recognize each virtual button press.

Method 2: Solving Hardware Domain Shift

The Problem: Models trained on known cards fail on unseen target cards.



Domain-Adversarial Architecture:

- The encoder first maps each RF signal into a 64-dimensional embedding. This embedding is sent to two branches: the label predictor and the domain discriminator.

Label Predictor:

- A classifier Multi-Layer Perceptron (MLP) maps the 64-dimensional embedding to the 9 virtual button classes.

Gradient Reversal Layer (GRL):

- The GRL is placed between the encoder and the domain discriminator. In forward propagation, it passes the embedding unchanged. In backpropagation, it multiplies the gradient by −λ, forcing the encoder to suppress card-specific features.

Domain Discriminator:

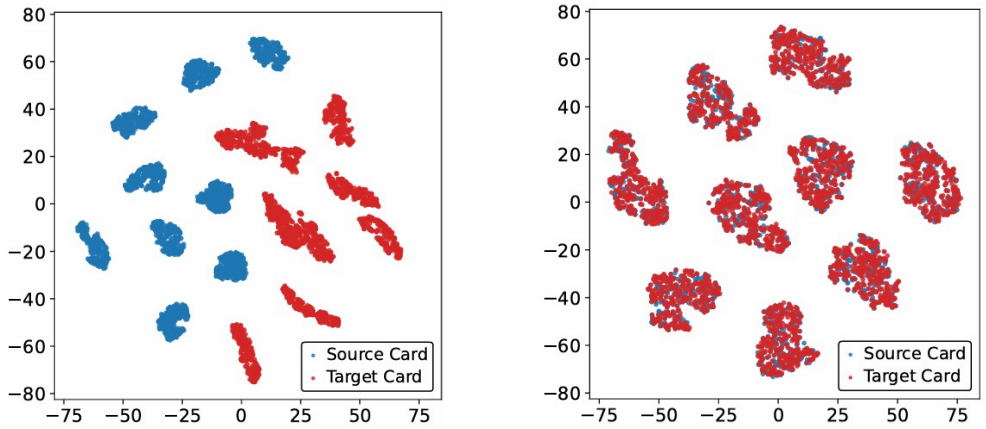
The discriminator is an MLP with a 128-unit fully connected layer, ReLU activation, dropout 0.5, a 64-unit fully connected layer, ReLU activation, and a final sigmoid output. It predicts whether the embedding comes from the source card or target card.

Result: The label predictor keeps button classes separable, while the GRL and discriminator make the embeddings card-invariant.

Cross-Card Prediction Performance

(a) before domain adaptation

(b) After domain adaptation



t-SNE visualization of encoder embeddings with and without domain adaptation for source and target NFC card button classes.

(a) Before Domain Adaptation

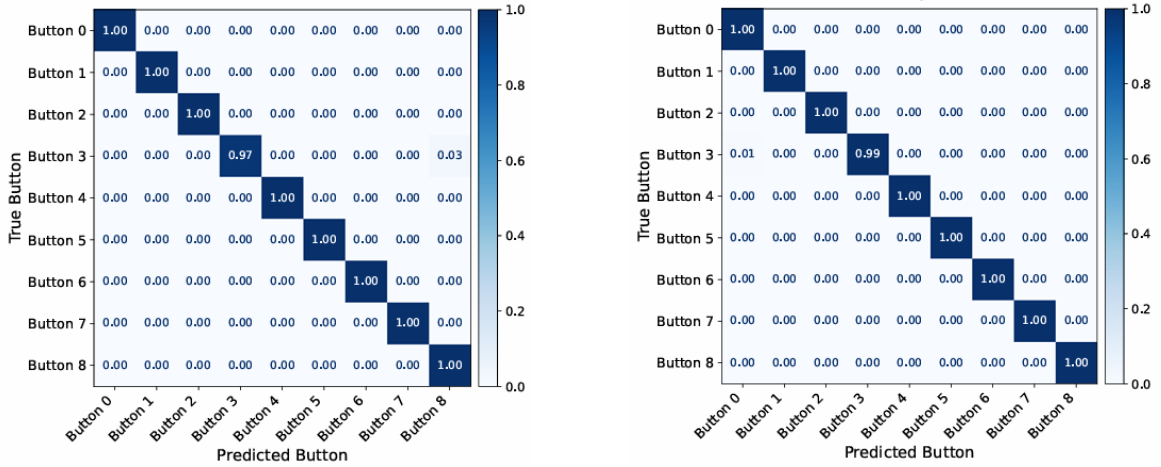
- Features separate mainly by NFC card hardware.
- Source and target embeddings form different clusters.

(b) After Domain Adaptation

- Source and target embeddings become well aligned.
- Features cluster by button class.

(a) Source card

(b) Target card



Confusion matrices for button class prediction using the adapted encoder.

Source prediction accuracy: 99.56% ± 0.69%

Target prediction accuracy: 99.72% ± 0.19%

- The frozen encoder successfully predicts buttons accurately across source and target cards.
- The system is now truly **Card-Agnostic.**

Method 3: Robust Recognition (Mahalanobis Distance)

The Problem:

- A traditional ML model always tries to assign every signal to one of the 9 buttons, even when the user presses outside the valid button areas. This can cause an invalid press to be wrongly accepted as a real PIN digit.

The Solution:

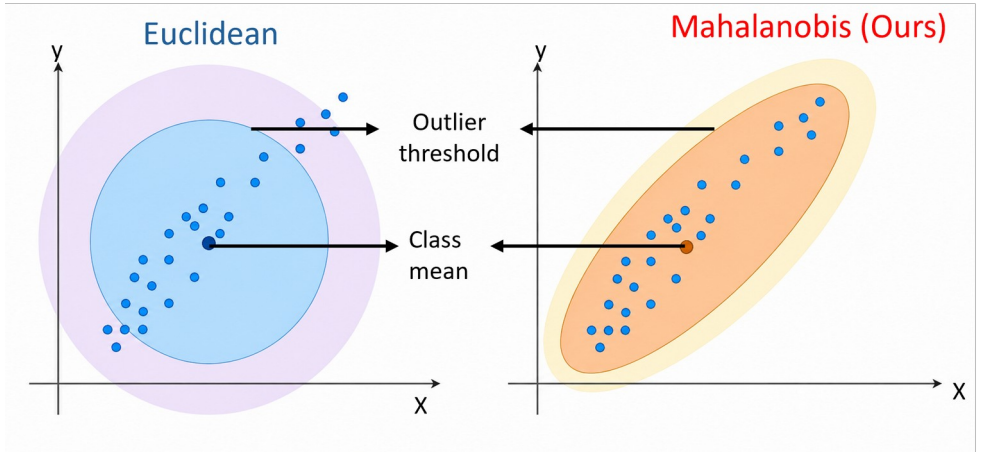
- Instead of trusting only the model's confidence score, we check how close the signal is to the normal pattern of each button in the learned feature space.

Why Mahalanobis Distance?

- Euclidean distance uses a fixed circular boundary, which does not match the real shape of the data. Mahalanobis distance is smarter because it follows the actual spread and direction of each button's signal cluster.

Result:

- If a signal falls inside the expected boundary, it is accepted as a valid button press. If it falls outside the boundary, it is rejected as noise, an invalid press, or a possible spoofed input.



Class Mean

$$u_c = \frac{1}{N_c} \sum_{x \in D_c^{\text{cal}}} f_\theta(x)$$

Squared Distance

$$d_i^2(x_i) = \big(f_\theta(x_i) - u_c\big)^\top S^{-1} \big(f_\theta(x_i) - u_c\big)$$

Cholesky Stable Form

Ensures viable $\mathcal{O}(Cd^2)$ computational stability.

$$d_i(x_i) = \big\| L^{-1} \big(f_\theta(x_i) - u_c\big) \big\|_2^2.$$

- u_c: center of genuine button class C
- $f_\theta(x_i)$: learned feature of the new press
- S^{-1}: adjusts distance based on how the button signals naturally spread.
- $d_i(x_i)$: distance from the new press to the expected button pattern.

Statistical Thresholding and Authentication Logic

Step 1: Build the "Impostor" Distance Set

We first measure how far incorrect button signals are from each valid button class. These distances represent invalid or non-matching button presses.

$$D_c^{\text{imp}} = \left\{ \left\| L^{-1}\left(f_\theta(x) - u_c\right) \right\|_2^2 : x \in D_q^{\text{cal}},\ q \neq c \right\}$$

Step 2: Choose a Decision Threshold

An α-quantile filter selects the acceptance threshold:

$$t_{c_i}(\alpha) = Q_\alpha\left(D_c^{\text{imp}}\right)$$

Step 3: Accept or Reject the Input

If the new button press is close enough to the expected button pattern, it is accepted. Otherwise, it is rejected.

$$\hat{a}(x_i, c_i; \alpha) = \begin{cases} 1, & \text{if } d_i(x_i) \leq t_{c_i}(\alpha), \\ 0, & \text{otherwise}, \end{cases}$$

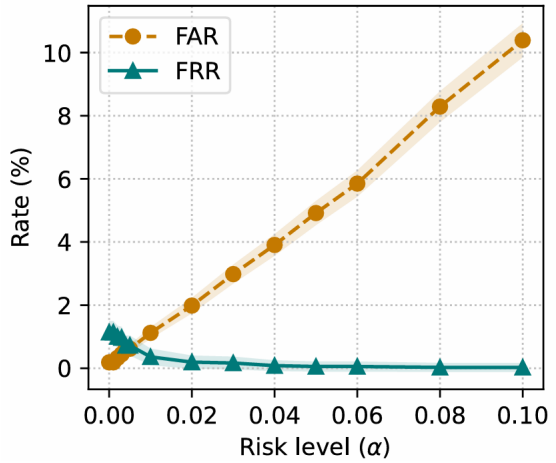
Step 4: Compute Recognition Acceptance Rate (AR)

After each test signal is classified and either accepted or rejected, we compute the final acceptance rate:

$$\text{AR} = \frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} \mathbb{I}[c_i = y_i \text{ and } \hat{a}(x_i, c_i; \alpha) = 1]$$

- N_{test}: total number of test samples
- c_i :predicted button class
- y_i :true button class
- $\hat{a}(\cdot)$:accept/reject decision
- AR measures how often the system both:
  - predicts the correct button, and
  - accepts the input as legitimate.

Impact of risk levels (α) on FAR and FRR



Security Balance: Tuning the α Risk Level

Low (α)	High (α)
Drives FAR severely low, increases FRR by rejecting more legitimate presses.	Drives FRR low but allows FAR to increase. The system becomes overly permissive, accepting more impostor distances.

Optimal Tuning: $\alpha = 0.025$

Provides a balanced FAR–FRR tradeoff across unseen NFC cards.

1.84% FAR vs 1.63% FRR across generalized domains.

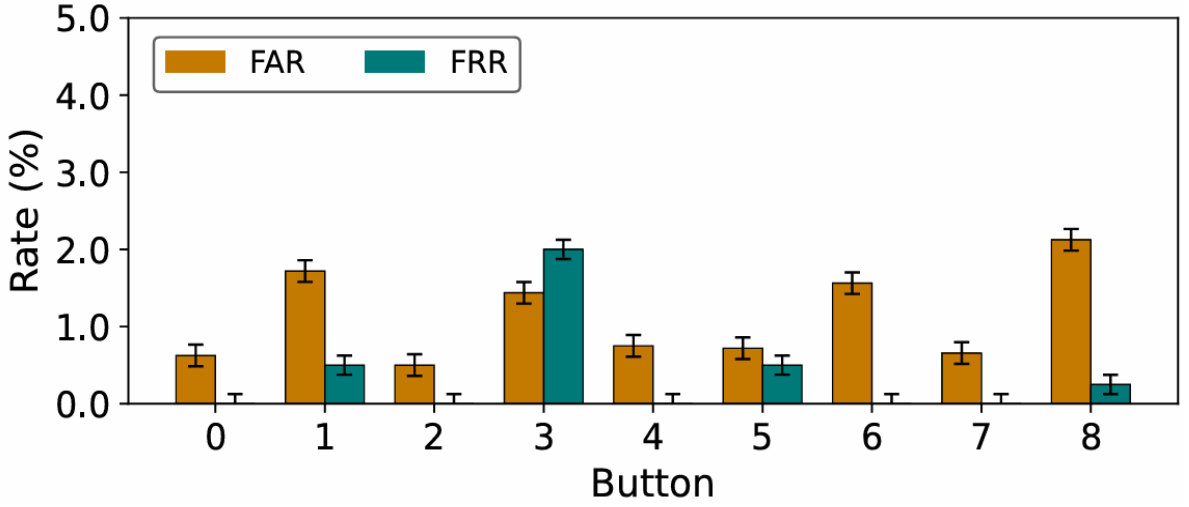
Results: Buttons Recognition on Unseen Cards

To evaluate generalization, we use a dataset from **12 unseen NXP ICODE SLIX (ISO/IEC 15693 SL2S2002) cards**, with each card providing 500 responses per button across all 9 virtual buttons, resulting in 54,000 test samples.

- **False Acceptance Rate (FAR):**
  1.84% (Incorrect responses accepted as genuine).
- **False Rejection Rate (FRR):**
  1.63% (Genuine responses incorrectly rejected).

- **Summary:**
  The system maintains stable, balanced sensitivity and specificity across all 9 virtual buttons.

98.20% Acceptance Rate
on completely new, 12 unseen NFC cards.



Average per-button FAR and FRR for unseen cards.

Results: Performance Under Noise

Experiment:

- Injecting AWGN (noise) to simulate severely degraded low SNR conditions.

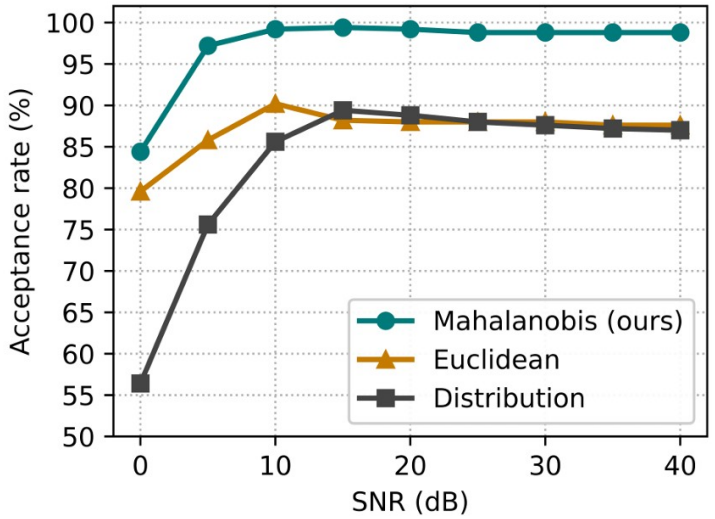
Findings:

- Baseline methods (Euclidean & Distribution) experience severe performance drops as noise increases.
- **CADRE (Mahalanobis)** maintains an Acceptance Rate of ***85%*** even in highly degraded conditions.

Conclusion:

- Covariance-aware geometric modeling provides superior real-world robustness.

Recognition performance using different methods



Algorithm Model	False Accept (FAR)	False Reject (FRR)	Acceptance Rate (AR)
Euclidean Distance	13.25%	12.60%	87.17%
Distribution Baseline	17.18%	16.33%	82.84%
Mahalanobis (Ours)	1.84%	1.63%	98.20%

Conclusion

- **CADRE introduces a new passive NFC authentication interface** that turns ordinary ISO/IEC 15693 NFC cards into virtual PIN pads without adding batteries, active electronics, or physical keypad hardware.
- **Unlike standard NFC, CADRE authenticates user interaction, not only card possession.** A valid card is no longer sufficient; the system also verifies the user's intended touch pattern through RF signal perturbations.
- **The proposed domain-aligned RF embedding framework overcomes card-to-card variations.** By suppressing hardware-specific distortions, CADRE generalizes reliably across unseen NFC cards instead of being limited to cards used during training.
- **Mahalanobis-based recognition strengthens security by rejecting invalid and uncertain inputs.** The system does not blindly classify every signal as a PIN digit; it rejects out-of-distribution, noisy, or spoof-like responses.
- **Overall, CADRE achieves over 98% authentication accuracy on unseen NFC cards while remaining low-cost, battery-free, passive, and compliant with ISO/IEC 15693 standards.**

Q/A

Thank you!

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